Identifying Key Entities in Recipe Data Using Conditional Random Fields (CRF)

Abstract

The objective of this project was to develop a Named Entity Recognition (NER) model capable of automatically identifying important entities from recipe ingredient lists. The entities of interest were **Quantity**, **Unit**, and **Ingredient**. Accurate extraction of these entities is an important preprocessing step for recipe recommendation systems, nutritional analysis, grocery list generation, and other food-related Natural Language Processing (NLP) applications.

A Conditional Random Fields (CRF) model was implemented because it is well suited for sequence labeling tasks where the prediction of one token depends on its surrounding context. The complete workflow included data understanding, preprocessing, feature engineering, model training, evaluation, prediction on unseen data, and detailed error analysis.

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1. Introduction

Recipe ingredients are generally written in free-text format and do not follow a fixed grammatical structure. A single ingredient line often contains quantities, measurement units, ingredient names, descriptive words, and cooking instructions. Before recipes can be analyzed computationally, these components need to be separated into meaningful entities.

Named Entity Recognition (NER) provides an effective solution by assigning a label to every token in the ingredient text. In this project, the following labels were identified:

- **Quantity** – Numeric values such as *1*, *2*, *500*, *½*.
- **Unit** – Measurement units such as *cup*, *tablespoon*, *grams*, *teaspoon*, etc.
- **Ingredient** – Actual ingredient names such as *rice*, *tomato*, *garlic*, *coriander*, and many others.

The project aims to build a machine learning model capable of automatically extracting these entities with high accuracy.

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2. Dataset Description

The dataset was provided in JSON format.

Each record contains two fields:

Input

The **input** field contains the raw ingredient sequence extracted from a recipe.

Example:

"2 cups rice 1 teaspoon cumin seeds"

POS Labels

The **pos** field stores the corresponding labels for every token present in the input sentence.

Example:

Token	Label
2	Quantity
cups	Unit
rice	Ingredient
1	Quantity
teaspoon	Unit
cumin	Ingredient
seeds	Ingredient

Since every token already has an associated label, the dataset is suitable for supervised sequence learning.

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3. Project Methodology

The project was completed in multiple stages.

3.1 Library Installation

The required Python libraries were installed before starting the implementation.

The major libraries used include:

- pandas
- numpy
- sklearn-crfsuite
- python-crfsuite
- spaCy
- matplotlib
- scikit-learn

These libraries were used for preprocessing, feature engineering, model development, visualization, and evaluation.

3.2 Data Loading

The JSON dataset was loaded into a pandas DataFrame.

The input ingredient strings and their corresponding entity labels were extracted and verified to ensure that every token had a matching label.

This validation step helped eliminate inconsistencies before model training.

3.3 Data Preprocessing

The dataset was converted into a sequence labeling format.

For every recipe:

- the ingredient sentence was tokenized,
- each token was aligned with its corresponding label,
- training instances were created as token-label pairs.

This conversion is necessary because CRF models learn from sequences rather than isolated words.

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4. Feature Engineering

The performance of a CRF model depends heavily on feature quality.

Instead of relying only on the current word, multiple contextual features were extracted.

The features included:

- Current token
- Lowercase form
- Prefixes
- Suffixes
- Word length
- Digit information
- Uppercase information
- Title case information
- Previous token
- Next token
- Beginning-of-sentence indicator
- End-of-sentence indicator

Using neighbouring words enables the model to understand contextual relationships between tokens.

For example:

Input:

2 cups rice

The model learns that:

- "2" is generally followed by a measurement unit.
- "cups" usually appears between a quantity and an ingredient.
- "rice" frequently appears after a measurement unit.

These contextual dependencies significantly improve prediction accuracy.

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5. Train–Validation Split

The dataset was divided into training and validation subsets.

The training data was used for learning model parameters, while the validation data was used to evaluate the model's generalization capability on unseen examples.

Separating the datasets prevents overfitting and provides an unbiased estimate of model performance.

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6. Conditional Random Fields (CRF) Model

Conditional Random Fields were selected because they are specifically designed for sequence prediction problems.

Unlike traditional classifiers that make independent predictions, CRFs consider relationships among neighbouring labels.

For example:

A quantity is generally followed by a measurement unit, which is then followed by an ingredient.

The model learns these transition patterns during training, making it highly effective for Named Entity Recognition tasks.

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7. Model Training

After feature extraction, the CRF classifier was trained using the processed training sequences.

The model learned both:

- relationships between token features and entity labels, and
- dependencies among consecutive labels.

This allows the model to predict complete sequences instead of individual words independently.

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8. Model Evaluation

The trained model was evaluated using standard classification metrics.

The evaluation included:

- Precision

- Recall
- F1-score
- Accuracy
- Confusion Matrix

These metrics provide a comprehensive assessment of model performance.

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9. Training Performance

The model achieved perfect performance on the training dataset.

Entity	Precision	Recall	F1-Score
Quantity	1.00	1.00	1.00
Unit	1.00	1.00	1.00
Ingredient	1.00	1.00	1.00

Overall training accuracy reached **100%**.

Although perfect training accuracy demonstrates that the model successfully learned the training patterns, validation performance provides a more reliable indication of real-world capability.

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10. Validation Performance

The model was then evaluated using the validation dataset.

The obtained results were:

Entity	Precision	Recall	F1-Score
Quantity	0.99	1.00	0.99
Unit	0.99	0.99	0.99
Ingredient	1.00	1.00	1.00

Overall validation accuracy remained approximately **100%**, indicating excellent generalization performance.

These results demonstrate that the CRF model successfully learned the structural patterns present in recipe ingredient lists without significant overfitting.

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11. Error Analysis

Although the validation performance was exceptionally high, a detailed error analysis was performed.

The notebook reported:

- **Total validation tokens:** 2876
- **Total prediction errors:** 5
- **Overall validation accuracy: 99.83%**

The error distribution showed:

True Label	Number of Errors	Accuracy
Unit	3	99.16%
Quantity	2	99.51%

No significant errors were observed for the Ingredient class.

The few mistakes mainly occurred in ambiguous tokens.

Examples include words such as **"to"** (as in "10 to 12") and **"into"** (as in "French cut into"), which appeared in contexts that resembled measurement expressions. Such tokens occasionally confused the model into predicting a Quantity label instead of a Unit label.

Despite these isolated cases, the overall error rate remained extremely low.

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12. Observations

Several important observations were made during the project.

- Ingredient names are generally easier to recognize because they occur frequently and have consistent contextual patterns.
- Quantities are highly distinguishable since they are mostly represented by numeric values.
- Units are sometimes more challenging because certain words can appear in different contexts depending on the sentence structure.
- Contextual features greatly improved prediction accuracy.
- CRF effectively captured dependencies between neighbouring tokens, reducing inconsistent predictions.

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13. Applications

The developed NER model can be integrated into several practical applications.

These include:

- Recipe recommendation systems
- Smart grocery list generation
- Nutrition information extraction

- Food knowledge graphs
- Recipe search engines
- Cooking assistants
- Meal planning applications
- Ingredient standardization pipelines

Automatic extraction of structured recipe information can significantly simplify downstream data analysis tasks.

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14. Limitations

Although the model performed exceptionally well, some limitations remain.

- The model is trained specifically on recipe ingredient data and may not generalize well to unrelated domains.
- Rare ingredient names or uncommon measurement expressions may reduce prediction accuracy.
- Highly irregular formatting or spelling mistakes may introduce prediction errors.
- Additional entity categories such as cooking methods, preparation instructions, temperatures, and durations are not recognized.

Future work could extend the model to identify these additional entities.

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15. Future Improvements

Several enhancements can further improve the system.

- Increase dataset size using recipes from multiple cuisines.
- Introduce additional entity labels such as cooking method and preparation style.
- Apply hyperparameter optimization for improved performance.
- Compare CRF with modern transformer-based architectures such as BERT.
- Incorporate contextual word embeddings.
- Perform cross-validation for more robust evaluation.

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16. Conclusion

This project successfully implemented a Named Entity Recognition system for extracting structured information from recipe ingredient lists using Conditional Random Fields.

The complete pipeline included data preprocessing, feature engineering, CRF model training, validation, prediction, and detailed error analysis. The model demonstrated excellent performance,

achieving perfect training scores and approximately 100% validation performance, with an overall validation accuracy of **99.83%** and only **5 prediction errors out of 2,876 validation tokens**.

The analysis further revealed that the few remaining errors occurred primarily in ambiguous contextual expressions rather than genuine ingredient names, indicating that the model learned meaningful linguistic patterns rather than relying solely on memorization.

Overall, the developed CRF-based NER system proved to be a highly reliable solution for extracting quantities, measurement units, and ingredient names from unstructured recipe text. The results demonstrate that Conditional Random Fields remain an effective and computationally efficient approach for structured prediction tasks, particularly when carefully designed contextual features are incorporated into the learning process.